FOURIER SERIES

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Inner – Product

» Inner product

- An inner product is a function that maps each ordered pairs of vectors $\boldsymbol{x}, \boldsymbol{y}$ to a real scalar, $\langle \boldsymbol{x} | \boldsymbol{y} \rangle$

$$\langle \boldsymbol{x} | \boldsymbol{y} \rangle = \boldsymbol{x}^{\mathrm{T}} \boldsymbol{y} \qquad \boldsymbol{x}, \boldsymbol{y} \in \mathrm{R}^{n \times 1}$$

$$= \sum_{i=1}^{n} x_i y_i$$

Two vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathrm{V}$ are said to be orthogonal (to each other) whenever $\langle \boldsymbol{x} | \boldsymbol{y} \rangle = 0$, and this is denoted by $\boldsymbol{x} \perp \boldsymbol{y}$

$$\boldsymbol{x} \perp \boldsymbol{y} \Leftrightarrow \boldsymbol{x}^{\mathrm{T}} \boldsymbol{y} = 0$$

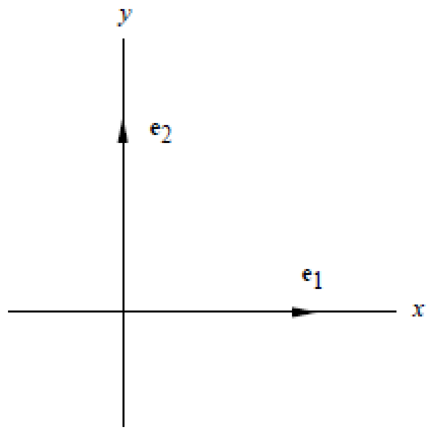
» Example

$$\boldsymbol{x} = \begin{pmatrix} 1 \\ -2 \\ 3 \\ -1 \end{pmatrix} \qquad \boldsymbol{y} = \begin{pmatrix} 4 \\ 1 \\ -2 \\ 4 \end{pmatrix}$$

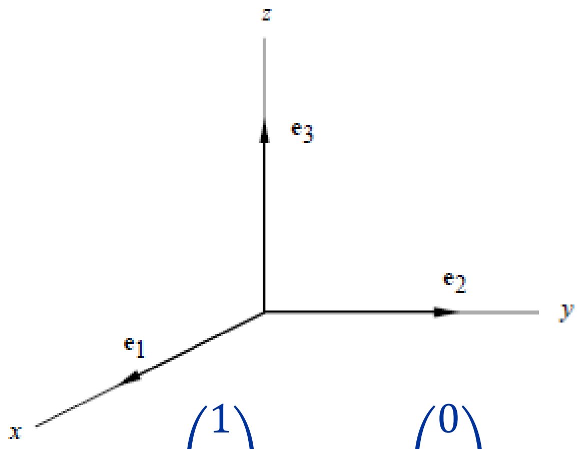
$$\boldsymbol{x}^{\mathrm{T}} \boldsymbol{y} = (1)(4) + (-2)(1) + (3)(-2) + (-1)(-4) = 0$$

$\Rightarrow \boldsymbol{x}, \boldsymbol{y}$ are orthogonal

Fourier Expansion



$$\boldsymbol{e}_1 = \begin{pmatrix}1\\0\end{pmatrix} \quad \boldsymbol{e}_2 = \begin{pmatrix}0\\1\end{pmatrix}$$



$$\boldsymbol{e}_1 = \begin{pmatrix}1\\0\\0\end{pmatrix} \quad \boldsymbol{e}_2 = \begin{pmatrix}0\\1\\0\end{pmatrix} \quad \boldsymbol{e}_3 = \begin{pmatrix}0\\0\\1\end{pmatrix}$$

$$\begin{aligned}\boldsymbol{x} &= \langle \boldsymbol{e_1}|\boldsymbol{x}\rangle \boldsymbol{e_1} + \langle \boldsymbol{e_2}|\boldsymbol{x}\rangle \boldsymbol{e_2} + \langle \boldsymbol{e_3}|\boldsymbol{x}\rangle \boldsymbol{e_3} \\ &= x_1\boldsymbol{e_1} + x_2\boldsymbol{e_2} + x_3\boldsymbol{e_3} \\ &= \begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}\end{aligned}$$

» Orthonormal sets

If B $= \{\boldsymbol{u}_1, \boldsymbol{u}_2, \dots, \boldsymbol{u}_n\}$ is an orthonormal basis for an inner – product space R^{n}, then each $\boldsymbol{x} \in \mathrm{R}^{\mathrm{n}}$ can be expressed as

$$\begin{aligned}\boldsymbol{x} &= \langle \boldsymbol{u_1}|\boldsymbol{x}\rangle \boldsymbol{u_1} + \langle \boldsymbol{u_2}|\boldsymbol{x}\rangle \boldsymbol{u_2} + \cdots + \langle \boldsymbol{u_n}|\boldsymbol{x}\rangle \boldsymbol{u_n} \\ &= \sum_{i=1}^{\infty} \xi_i \boldsymbol{u_i}\end{aligned}$$

This is called the Fourier expansion of $\boldsymbol{x}$. The scalars $\xi_i = \langle \boldsymbol{u_i}|\boldsymbol{x}\rangle$ are the coordinates of $\boldsymbol{x}$ with respect to B , and they are called the Fourier coefficients.

Fourier Expansion

» Inner product of functions

- Think of function as a vector in an infinite – dimensional space

  Consider real – valued continuous functions, $f(x)$ and $g\,(x)$defined on the interval (a, b),

  $$\langle f|g\rangle = \int_a^b f(x)g(x)dt$$

  Two functions $f,\ g$ are said to be orthogonal (to each other) whenever,

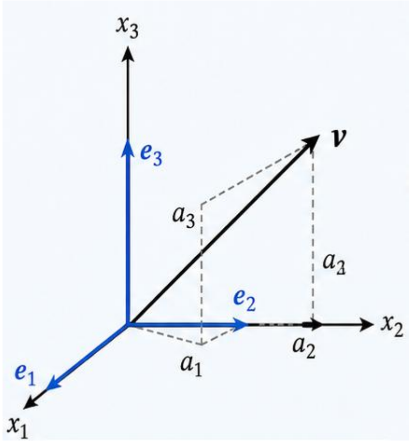
  $$\langle f|g\rangle = 0$$

- Let functions $\phi_1(x), \phi_2(x), \ldots$ defined on some $a \le x \le b$ be orthonormal

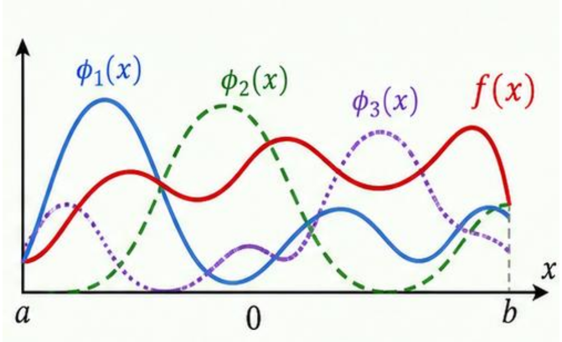
  $$f(x) = \langle \phi_1|f\rangle\phi_1(x) + \langle \phi_2|f\rangle\phi_2(x) + \cdots + \langle \phi_n|f\rangle\phi_n(x) + \ \ldots$$

  $$= a_1\phi_1(x) + a_2\phi_2(x) + \cdots$$

  $$= \sum_{i=1}^{\infty} a_i\phi_i(x)$$



$$\boldsymbol{v} = a_1\boldsymbol{e_1} + a_2\boldsymbol{e_2} + a_3\boldsymbol{e_3}$$

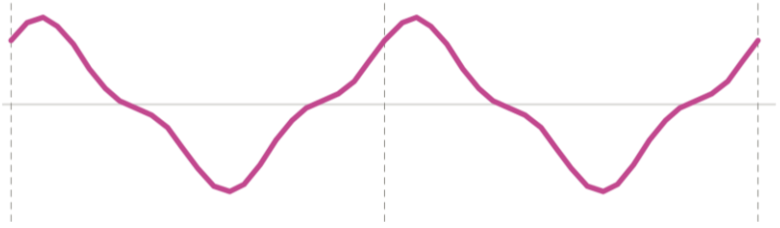


$$f(x) = a_1\phi_1(x) + a_2\phi_2(x) + a_3\phi_3(x)$$

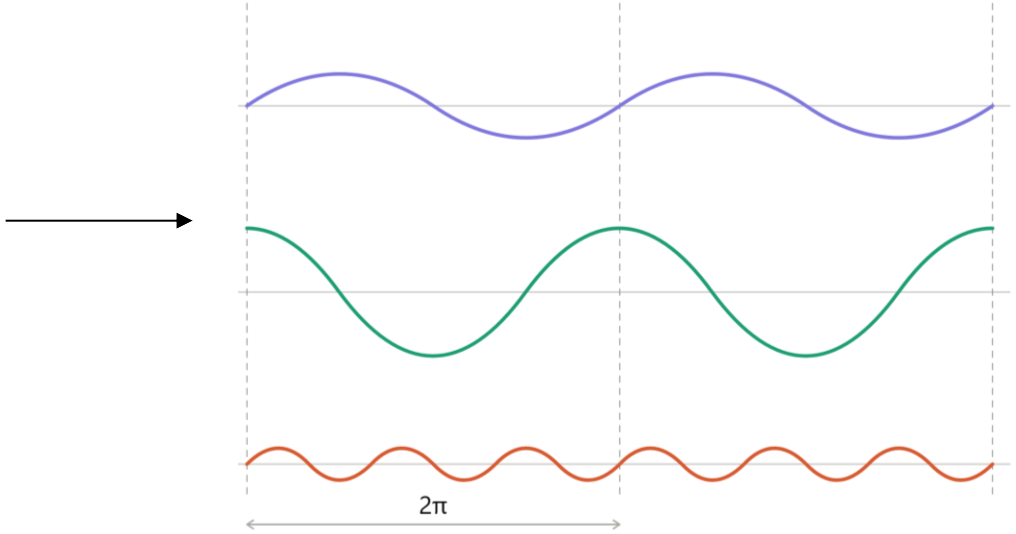
Periodic Functions

» Fourier series: Idea

- Periodic signals as sums of harmonics



$f(x + p) = f(x)$ for all x



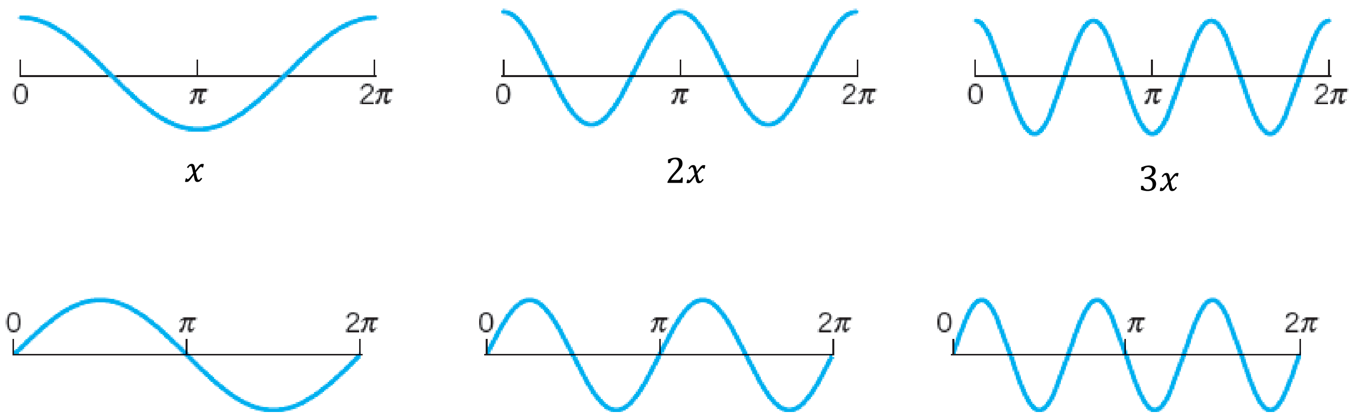
Periodic Functions

» Trigonometric system

$1, \quad \cos x, \sin x, \quad \cos 2x, \sin 2x, \quad \dots, \cos nx, \sin nx$

- Trigonometric functions are orthogonal

$$\int_{-\pi}^{\pi} \sin nx \cos mx \; dx = 0, \qquad (n \neq m, \qquad n = m)$$



(Top) Cosine and (bottom) sine functions having period 2π

$$\int_{-\pi}^{\pi} \sin nx \cos mx \; dx = \frac{1}{2}\int_{-\pi}^{\pi} [\sin(n+m)x \; dx + \sin(n+m)x] \; dx$$

$$= \frac{1}{2}\left(-\frac{\cos\,(n+m)x}{n+m}\Big|_{-\pi}^{\pi} - \frac{\mathrm{coss}\,(n-m)x}{n-m}\Big|_{-\pi}^{\pi} \right)$$

$$= 0$$

Periodic Functions

» Cosine family

$1, \quad \cos x\,, \quad \cos 2x\,, \quad \ldots, \cos nx$

$$\boxed{\int_{-\pi}^{\pi} \cos nx \cos mx \ dx = 0\,, \qquad (n \neq m)}$$

$$\int_{-\pi}^{\pi} \cos nx \cos mx \ dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi} [\cos(n+m)x \ dx + \cos(n-m)x] \ dx$$

$$= \frac{1}{2}\left(\frac{\sin\,(n+m)x}{n+m}\bigg|_{-\pi}^{\pi} + \frac{\sin\,(n-m)x}{n-m}\bigg|_{-\pi}^{\pi} \right)$$

$$= 0$$

» Sine family

$1, \quad \sin x\,, \quad \sin 2x\,, \quad \ldots, \sin nx$

$$\boxed{\int_{-\pi}^{\pi} \sin nx \sin mx \ dx = 0\,, \qquad (n \neq m)}$$

$$\int_{-\pi}^{\pi} \sin nx \sin mx \ dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi} [\cos(n+m)x \ dx - \cos(n-m)x] \ dx$$

$$= \frac{1}{2}\left(\frac{\sin\,(n-m)x}{n-m}\bigg|_{-\pi}^{\pi} + \frac{\sin\,(n+m)x}{n+m}\bigg|_{-\pi}^{\pi} \right)$$

$$= 0$$

Fourier Series

» Fourier series

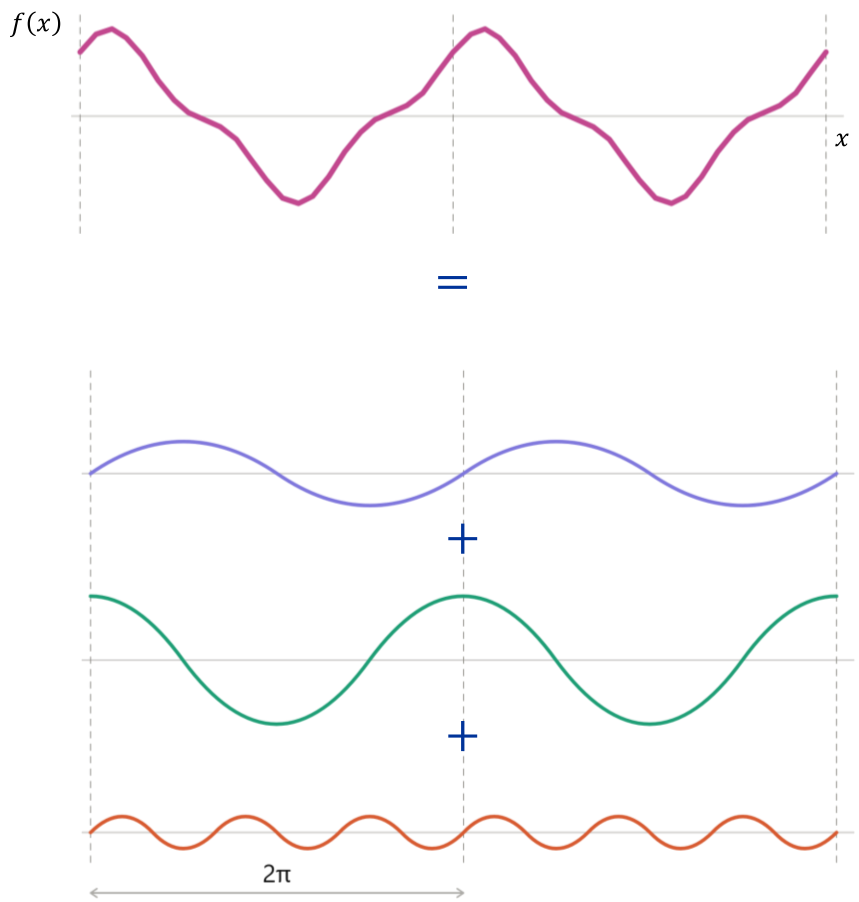
- Simple functions having period, $\mathrm{p} = 2\pi$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\, dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \cos nx\, dx\,, \qquad n = 1, 2, \ldots$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx\, dx\,, \qquad n = 1, 2, \ldots$$



Fourier Series

» Determination of Fourier coefficients

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$\int_{-\pi}^{\pi} f(x)\,dx = \int_{-\pi}^{\pi} \left[a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \right] dx$$

$$= \int_{-\pi}^{\pi} a_0 dx + \sum_{n=1}^{\infty} [a_n \int_{-\pi}^{\pi} \cos nx\,dx + b_n \int_{-\pi}^{\pi} \sin nx\,dx]$$

$$= 2\pi a_0 + \sum_{n=1}^{\infty} [a_n \frac{\sin\ nx}{n}\Big|_{-\pi}^{\pi} + b_n \frac{\cos\ nx}{n}\Big|_{-\pi}^{\pi}]$$

$$= 2\pi a_0$$

$$\boxed{a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\,dx}$$

Fourier Series

$$\int_{-\pi}^{\pi} f(x)\cos mx\,dx = \int_{-\pi}^{\pi}\left[a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx\right]\cos mx\,dx$$

$$= a_0 \int_{-\pi}^{\pi} \cos mx\,dx + \sum_{n=1}^{\infty}\left[a_n \int_{-\pi}^{\pi} \cos nx \cos mx\,dx + b_n \int_{-\pi}^{\pi} \sin nx \cos mx\,dx\right]$$

$$= 0, \qquad m \neq n$$

$$\int_{-\pi}^{\pi} f(x)\cos nx\,dx = a_n \int_{-\pi}^{\pi} \cos^2 nx\,dx \qquad m = n$$

$$= \frac{a_n}{2}\int_{-\pi}^{\pi} (1 + \cos 2nx)\ dx$$

$$= \frac{a_n}{2}\left[\int_{-\pi}^{\pi} dx + \int_{-\pi}^{\pi} \cos 2nx\ dx\right]$$

$$= \frac{a_n}{2}\left[(2\pi) + \frac{\sin\ 2nx}{2n}\Big|_{-\pi}^{\pi}\right]$$

$$\boxed{a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx\,, \qquad n = 1, 2, \ldots}$$

Fourier Series

$$\int_{-\pi}^{\pi} f(x)\sin mx\,dx = \int_{-\pi}^{\pi}\left[a_0 + \sum_{n=1}^{\infty} a_n\cos nx + b_n\sin nx\right]\sin mx\,dx$$

$$= a_0\int_{-\pi}^{\pi}\sin mx\,dx + \sum_{n=1}^{\infty}[a_n\int_{-\pi}^{\pi}\cos nx\sin mx\,dx + b_n\int_{-\pi}^{\pi}\sin nx\sin mx\,dx]$$

$$= 0, \qquad m \neq n$$

$$\int_{-\pi}^{\pi} f(x)\sin nx\,dx = b_n\int_{-\pi}^{\pi}\sin^2 nx\,dx \qquad m = n$$

$$= \frac{b_n}{2}\int_{-\pi}^{\pi}(1-\cos 2nx)\ dx$$

$$= \frac{b_n}{2}\left[\int_{-\pi}^{\pi}dx - \int_{-\pi}^{\pi}\cos 2nx\ dx\right]$$

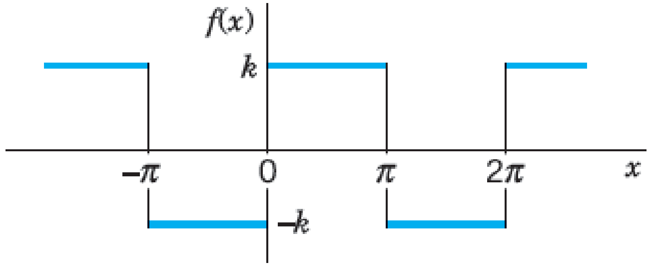
$$= \frac{b_n}{2}\left[(2\pi) - \frac{\sin\ 2nx}{2n}\bigg|_{-\pi}^{\pi}\right]$$

$$\boxed{b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx\,, \qquad n = 1, 2, \ldots}$$

Fourier Series

» Example

$$f(x) = \begin{cases} -k, & if \ -\pi < x < 0 \\ k, & if \ 0 < x < \pi \end{cases}$$



Periodic rectangular wave

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$
$$= \frac{1}{2\pi}\left(\int_{-\pi}^{0}(-k)dx + \int_{0}^{\pi}(k)dx\right)$$
$$= \frac{1}{2\pi}k(-\pi + \pi) = 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$
$$= \frac{1}{\pi}\left(\int_{-\pi}^{0}(-k)\cos nx\,dx + \int_{0}^{\pi}(k)\cos nx\,dx\right)$$
$$= \frac{k}{\pi}\left(-\frac{\sin nx}{n}\Big|_{-\pi}^{0} + \frac{\sin nx}{n}\Big|_{0}^{\pi}\right)$$
$$= 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$
$$= \frac{1}{\pi}\left(\int_{-\pi}^{0}(-k)\sin nxdx + \int_{0}^{\pi}(k)\sin nxdx\right)$$
$$= \frac{k}{\pi}\left(\frac{\cos nx}{n}\Big|_{-\pi}^{0} - \frac{\cos nx}{n}\Big|_{0}^{\pi}\right)$$
$$= \frac{k}{n\pi}\left[(1 - \cos n\pi) - (\cos n\pi - 1)\right]$$
$$= \frac{2k}{n\pi}(1 - \cos n\pi)$$
$$b_n = \begin{cases} \dfrac{4k}{n\pi}, & n \ is \ \text{odd} \\ 0, & n \ is \ \text{even} \end{cases}$$

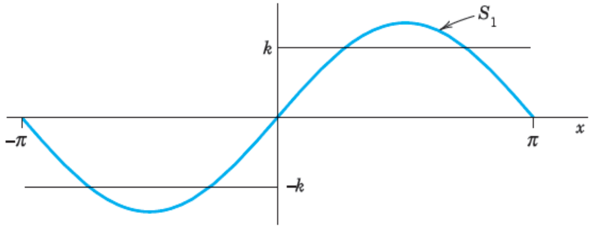
Fourier Series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

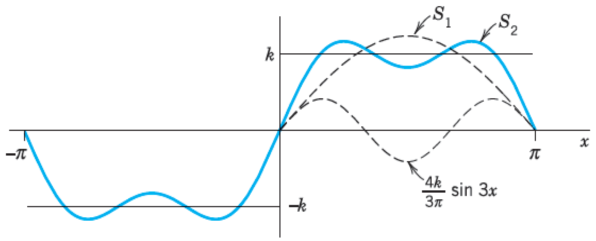
$$= \sum_{n=1}^{\infty} b_n \sin nx = \frac{4k}{\pi}\left(\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \ \dots\right)$$

```
% Periodic Rectangular Wave
k = 3;
x = -pi:0.1:pi;
s = zeros(1, size(x, 2));
for n = 1:3
    % Fourier Coefficient
    b = (2*k/(n*pi))*(1 - cos(n*pi));
    f = b*sin(n*x);
    s = s+f; % nth partial sum
end
plot(x,s)
```

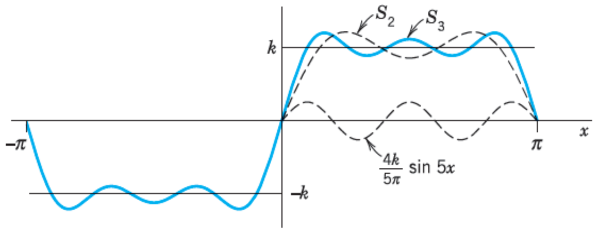
$$S_1 = \frac{4k}{\pi}\sin x$$



$$S_2 = \frac{4k}{\pi}\ \left(\sin x + \frac{1}{3}\sin 3x\right)$$



$$S_3 = \frac{4k}{\pi}\ \left(\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x\right)$$



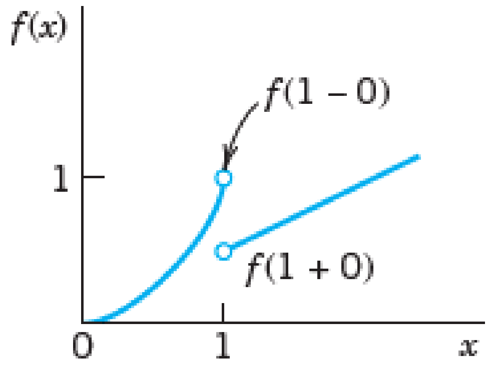
Fourier Series

» Fourier series: Convergence

- Let $f(x)$ be periodic with period 2π, and piecewise continuous, then the Fourier series of function $f(x)$ converges.
- For points where $f(x)$ is discontinuous (say, x_0) , the sum of series is the average of the left- and right-hand limits of $f(x)$ at x_0

» Example

$$f(x) = \begin{cases} x^2, & if\ x < 1 \\ \dfrac{x}{2} & if\ x \geq 1 \end{cases}$$



$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} x^2 = 1$$

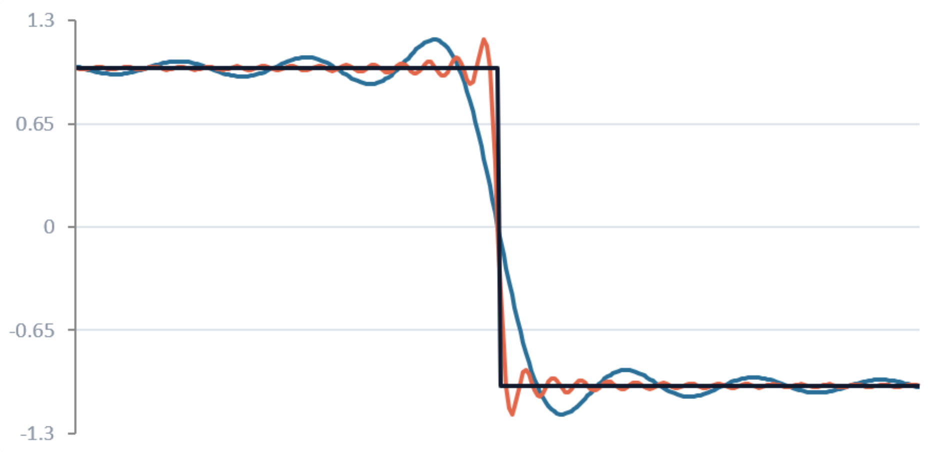
$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} \frac{x}{2} = \frac{1}{2}$$

$$S(1) = \frac{\lim_{x \to 1^-} f(x) + \lim_{x \to 1^+} f(x)}{2} = \frac{3}{4}$$

Fourier Series

» Fourier series: Gibbs phenomenon

- Gibbs phenomenon is the oscillation that appears near a jump discontinuity
- More Fourier terms make the oscillations narrower, but not eliminate the peak overshoot



Fourier Series

» Change of period

- Simple functions having period, $\mathrm{p} = 2\mathrm{L}$

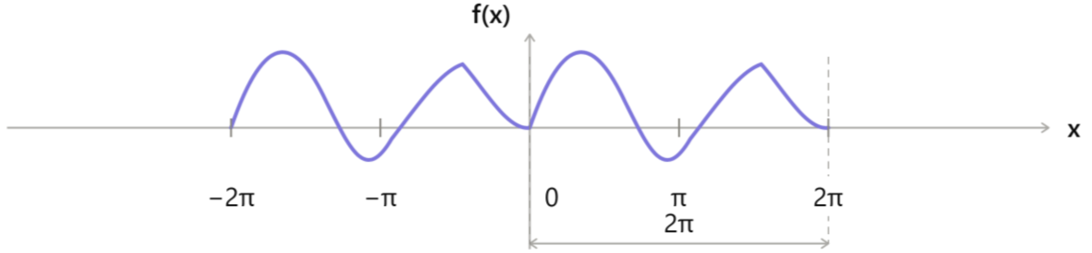
$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

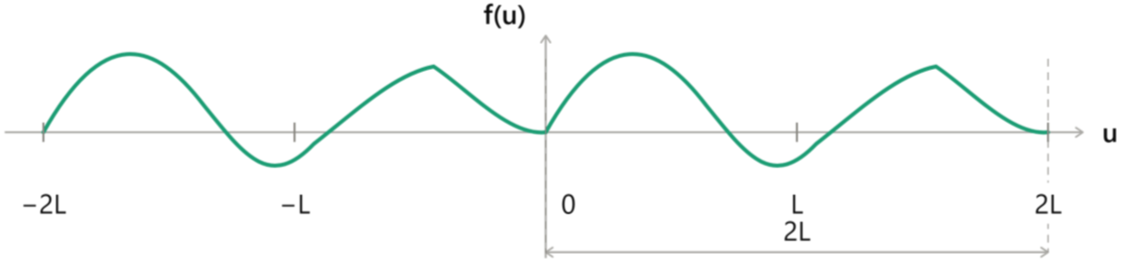
$$f(u) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\frac{\pi}{\mathrm{L}}u + b_n \sin n\frac{\pi}{\mathrm{L}}u$$

$$\boxed{a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\, dx}$$

$$a_0 = \frac{1}{2\pi}\int_{-\mathrm{L}}^{\mathrm{L}} f(u)\frac{\pi}{\mathrm{L}} du$$

$$= \frac{1}{2\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}} f(u)\, du$$





$$\frac{x}{u} = \frac{2\pi}{2\mathrm{L}} \quad \rightarrow \quad x = \frac{\pi}{\mathrm{L}}u$$

Fourier Series

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx, \qquad n = 1, 2, \ldots$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx, \qquad n = 1, 2, \ldots$$

$$a_n = \frac{1}{\pi}\int_{-\mathrm{L}}^{\mathrm{L}} f(u)\cos\left(\frac{n\pi}{\mathrm{L}}u\right)\frac{\pi}{\mathrm{L}}udu$$

$$= \frac{1}{\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}} f(u)\cos\left(\frac{n\pi}{\mathrm{L}}u\right)du$$

$$b_n = \frac{1}{\pi}\int_{-\mathrm{L}}^{\mathrm{L}} f(u)\sin\left(\frac{n\pi}{\mathrm{L}}u\right)\frac{\pi}{\mathrm{L}}udu$$

$$= \frac{1}{\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}} f(u)\sin\left(\frac{n\pi}{\mathrm{L}}u\right)du$$

» Fourier series

- Simple functions having period, $2\mathrm{L}$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n\cos\left(\frac{n\pi}{\mathrm{L}}x\right) + b_n\sin\left(\frac{n\pi}{\mathrm{L}}x\right)$$

$$a_0 = \frac{1}{2\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}} f(x)\,dx$$

$$a_n = \frac{1}{\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}} f(x)\cos\left(\frac{n\pi}{\mathrm{L}}x\right)dx, \qquad n = 1,2, \ldots$$

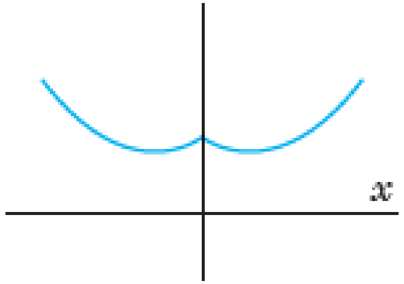
$$b_n = \frac{1}{\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}} f(x)\sin\left(\frac{n\pi}{\mathrm{L}}x\right)dx, \qquad n = 1,2, \ldots$$

Fourier Series: Even Function

» Even Function

$f(-x) = f(x)$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right)$$



Even function

$$\boxed{a_0 = \frac{1}{L}\int_0^L f(x)\,dx\,, \qquad a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi}{L}x\right)dx\,, \qquad n = 1, 2}$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi}{L}x\right)dx$$

$$g(x) = f(x)\cos\left(\frac{n\pi}{L}x\right)$$

$$g(-x) = f(-x)\cos\left(\frac{n\pi}{L}(-x)\right) = f(x)\cos\left(\frac{n\pi}{L}x\right) = g(x)$$

$$a_n = \frac{1}{L}\int_{-L}^{L} g(x)dx = \frac{2}{L}\int_0^L g(x)dx$$

$$\therefore\ a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi}{L}x\right)dx$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi}{L}x\right)dx\,, \qquad n = 1, 2$$

$$h(x) = f(x)\sin\left(\frac{n\pi}{L}x\right)$$

$$h(-x) = f(-x)\sin\left(\frac{n\pi}{L}(-x)\right) = -f(x)\sin\left(\frac{n\pi}{L}x\right) = -h(x)$$

$$\therefore\ b_n = \frac{1}{L}\int_{-L}^{L} h(x)dx = 0$$

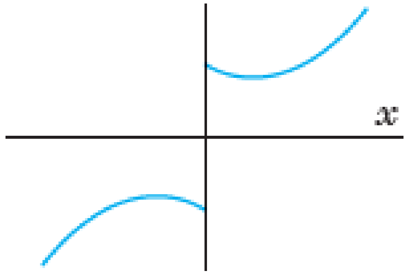
Fourier Series: Odd Function

» Odd Function

$$f(-x) = -f(x)$$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{\mathrm{L}}x\right)$$

$$\boxed{b_n = \frac{2}{\mathrm{L}}\int_0^{\mathrm{L}} f(x)\sin\left(\frac{n\pi}{\mathrm{L}}x\right)dx\,, \qquad n = 1, 2}$$



Odd function

$$a_0 = \frac{1}{2\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}} f(x)\,dx = 0$$

$$a_n = \frac{1}{\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}} f(x)\cos\left(\frac{n\pi}{\mathrm{L}}x\right)dx$$

$$g(x) = f(x)\cos\left(\frac{n\pi}{\mathrm{L}}x\right)$$

$$g(-x) = f(-x)\cos\left(\frac{n\pi}{\mathrm{L}}(-x)\right) = -f(x)\cos\left(\frac{n\pi}{\mathrm{L}}x\right) = -g(x)$$

$$\therefore a_n = \frac{1}{\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}} g(x)dx = 0$$

$$b_n = \frac{1}{\mathrm{L}}\int_{-\mathrm{L}}^{L} f(x)\sin\left(\frac{n\pi}{\mathrm{L}}x\right)dx\,, \qquad n = 1, 2$$

$$h(x) = f(x)\sin\left(\frac{n\pi}{\mathrm{L}}x\right)$$

$$h(-x) = f(-x)\sin\left(\frac{n\pi}{\mathrm{L}}(-x)\right) = f(x)\sin\left(\frac{n\pi}{\mathrm{L}}x\right) = h(x)$$

$$b_n = \frac{1}{\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}} g(x)dx = \frac{2}{\mathrm{L}}\int_0^{\mathrm{L}} h(x)dx$$

$$\therefore b_n = \frac{2}{\mathrm{L}}\int_0^{\mathrm{L}} f(x)\sin\left(\frac{n\pi}{\mathrm{L}}x\right)dx$$

Fourier Series : Theorem

» Sum and scalar multiple

- The Fourier coefficients of a sum, $f_1 + f_2$ are the sum of the corresponding Fourier coefficients of f_1 and f_2
- The Fourier coefficients of a of a scalar multiplication, cf are c time the Fourier coefficients of f

$$(cf_1+f_2)(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{\mathrm{L}}x\right) + b_n \sin\left(\frac{n\pi}{\mathrm{L}}x\right)$$

$$a_0 = \frac{1}{2\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}}(cf_1+f_2)dx \quad = \frac{1}{2\mathrm{L}}\left[\int_{-\mathrm{L}}^{\mathrm{L}} cf_1\, dx + \int_{-\mathrm{L}}^{\mathrm{L}} f_2\, dx\right] = ca_{01} + a_{02}$$

$$a_n = \frac{1}{\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}}(cf_1+f_2)\cos\left(\frac{n\pi}{\mathrm{L}}x\right)dx \quad = \frac{1}{2\mathrm{L}}\left[\int_{-\mathrm{L}}^{\mathrm{L}} cf_1 \cos\left(\frac{n\pi}{\mathrm{L}}x\right)dx + \int_{-\mathrm{L}}^{\mathrm{L}} f_2 \cos\left(\frac{n\pi}{\mathrm{L}}x\right)dx\right] = ca_{n1} + a_{n2}$$

$$b_n = \frac{1}{\mathrm{L}}\int_{-\mathrm{L}}^{\mathrm{L}}(cf_1+f_2)\sin\left(\frac{n\pi}{\mathrm{L}}x\right)dx \quad = \frac{1}{2\mathrm{L}}\left[\int_{-\mathrm{L}}^{\mathrm{L}} cf_1 \sin\left(\frac{n\pi}{\mathrm{L}}x\right)dx + \int_{-\mathrm{L}}^{\mathrm{L}} f_2 \sin\left(\frac{n\pi}{\mathrm{L}}x\right)dx\right] = cb_{n1} + b_{n2}$$

Fourier Series: Complex Form

» Fourier series

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

- Using Euler's Formula

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\therefore f(x) = a_0 + \sum_{n=1}^{\infty} a_n \left(\frac{e^{inx} + e^{-inx}}{2}\right) + b_n \left(\frac{e^{inx} + e^{-inx}}{2i}\right)$$

$$= a_0 + \sum_{n=1}^{\infty} \left(\frac{a_n - ib_n}{2}\right) e^{inx} + \left(\frac{a_n + ib_n}{2}\right) e^{-inx}$$

$$= c_0 + \sum_{n=1}^{\infty} c_n e^{inx} + \sum_{n=1}^{\infty} c_{-n} e^{-inx}$$

$$c_0 = a_0, \qquad c_n = \left(\frac{a_n - ib_n}{2}\right), \qquad c_{-n} = \left(\frac{a_n + ib_n}{2}\right)$$

Fourier Series: Complex Form

$$c_0 = a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)dx$$

$$c_n = \left(\frac{a_n - ib_n}{2}\right) = \frac{1}{2}\left[\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\cos nx\, dx \; - i\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\sin nx\, dx\right]$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\left(\frac{\cos nx - i\sin nx}{2}\right)dx$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}dx$$

$$c_{-n} = \left(\frac{a_n + ib_n}{2}\right) = \frac{1}{2}\left[\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\cos nx\, dx + i\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\sin nx\, dx\right]$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\left(\frac{\cos nx + i\sin nx}{2}\right)dx$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{inx}dx$$

$$\boxed{f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}, \qquad c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}dx, \qquad n = 0, \underline{+}1, \underline{+}2, \ldots}$$

Fourier Series: Time – Periodic

» Fourier series: Time - periodic signal

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2n\pi}{\mathrm{T}}t\right) + b_n \sin\left(\frac{2n\pi}{\mathrm{T}}t\right) \qquad \omega = \frac{2\pi}{\mathrm{T}}$$

$$= a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega t) + b_n \sin(n\omega t)$$

$$\boxed{a_0 = \frac{1}{\mathrm{T}}\int_{-\frac{\mathrm{T}}{2}}^{\frac{\mathrm{T}}{2}} f(t)\,dt}$$

$$\boxed{a_n = \frac{1}{\mathrm{T}}\int_{-\frac{\mathrm{T}}{2}}^{\frac{\mathrm{T}}{2}} f(t)\cos(n\omega t)\,dt}$$

$$\boxed{b_n = \frac{1}{\mathrm{T}}\int_{-\frac{\mathrm{T}}{2}}^{\frac{\mathrm{T}}{2}} f(t)\sin(n\omega t)\,dt}$$

» Time - periodic signal: Complex form

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega t}$$

$$\boxed{c_n = \frac{1}{\mathrm{T}}\int_{-\frac{\mathrm{T}}{2}}^{\frac{\mathrm{T}}{2}} f(t)e^{-in\omega t}\,dt, \qquad n = 0, \underline{+}1, \underline{+}2, \ldots}$$

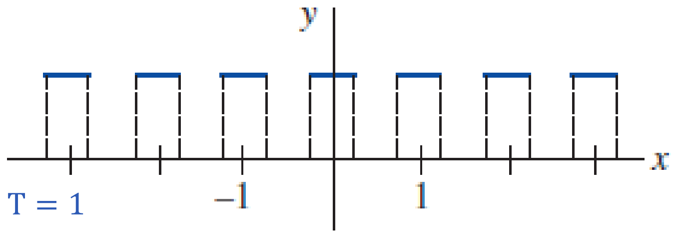
Fourier Series: Spectrum

» Fourier series: Frequency spectrum

- Using If $f(t)$ is periodic and has fundamental period T, the plot of the points $(n\omega, |c_n|)$, where ω is the fundamental angular frequency and the c_n are the coefficients, is called the **frequency spectrum** of $f(t)$.

» Example

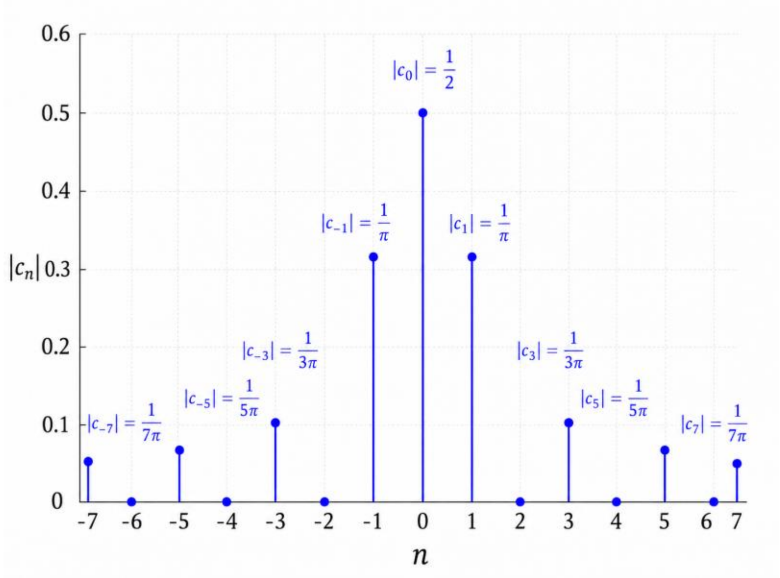
$$f(x) = \begin{cases} 0 & -\frac{1}{2} < x < -\frac{1}{4} \\ 1 & -\frac{1}{4} < x < \frac{1}{4} \\ 0 & \frac{1}{4} < x < \frac{1}{2} \end{cases}$$



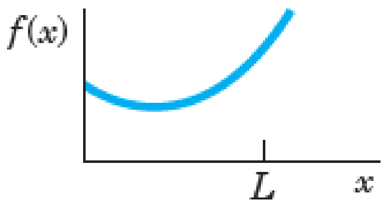
$$c_n = \int_{-\frac{1}{2}}^{\frac{1}{2}} f(t) e^{-2in\pi x} dx \quad = \int_{-\frac{1}{4}}^{\frac{1}{4}} e^{-2in\pi x} dx = -\frac{e^{-2in\pi x}}{2in\pi}\bigg|_{-\frac{1}{4}}^{\frac{1}{4}}$$

$$= -\frac{1}{n\pi}\left(\frac{e^{-\frac{in\pi}{2}} - e^{\frac{in\pi}{2}}}{2i}\right)$$

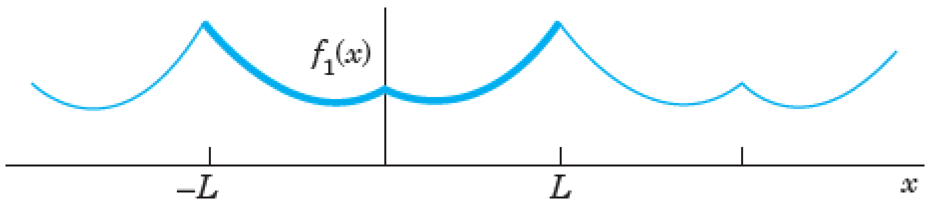
$$= \frac{1}{n\pi}\sin\frac{n\pi}{2}, \qquad n \neq 0$$



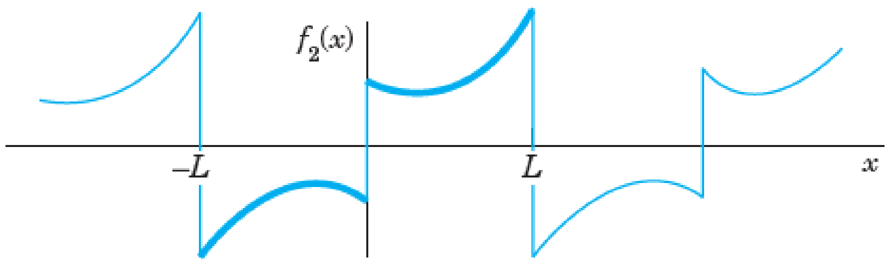
- Half – range expansions
  - Consider a function $f(t)$ defined on the range [0, L]
  - Example: Temperature along a bar, shape of a plucked string



Given function, $f(x)$



$f(x)$ continued as an even periodic function of period 2L



$f(x)$ continued as an odd periodic function of period 2L

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{\mathrm{L}} x\right)$$

$$a_0 = \frac{1}{\mathrm{L}} \int_0^{\mathrm{L}} f(x)\, dx\,, \qquad a_n = \frac{2}{\mathrm{L}} \int_0^{\mathrm{L}} f(x) \cos\left(\frac{n\pi}{\mathrm{L}} x\right) dx\,, \qquad n = 1, 2$$

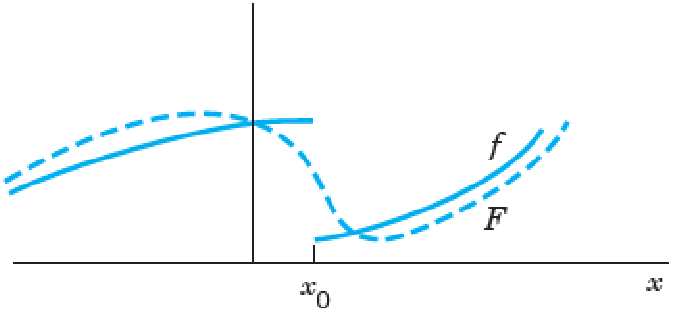
$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{\mathrm{L}} x\right)$$

$$b_n = \frac{2}{\mathrm{L}} \int_0^{\mathrm{L}} f(x) \sin\left(\frac{n\pi}{\mathrm{L}} x\right) dx\,, \qquad n = 1, 2$$

Function Approximation

» Fourier series: Function approximation

- Let $f(x)$ be a function on the interval $-\pi \leq x \leq \pi$ represented by a Fourier series



Error of approximation

- Approximation

$$F(x) = a_0 + \sum_{n=1}^{N} a_n \cos nx + b_n \sin nx$$

- Squared – error function

$$E = \int_{-\pi}^{\pi} (f - F)^2 \, dx = \int_{-\pi}^{\pi} (f^2 - 2fF + F^2) \, dx$$

$$\therefore \int_{-\pi}^{\pi} 2fF dx = \int_{-\pi}^{\pi} 2f \left(a_0 + \sum_{n=1}^{N} a_n \cos nx + b_n \sin nx \right) dx \quad = 2a_0 \int_{-\pi}^{\pi} f dx + 2 \sum_{i=1}^{N} \left(a_n \int_{-\pi}^{\pi} f \cos nx dx + b_n \int_{-\pi}^{\pi} f \sin nx dx \right)$$

$$= 2a_0(2\pi a_0) + 2 \sum_{i=1}^{N} \left(a_n(\pi a_n) + b_n(\pi b_n) \right)$$

$$= 4\pi a_0^2 + 2\pi \sum_{i=1}^{N} (a_n^2 + b_n^2)$$

Function Approximation

$$\therefore \int_{-\pi}^{\pi} \mathrm{F}^2 dx = \int_{-\pi}^{\pi} \left(a_0 + \sum_{n=1}^{\mathrm{N}} a_n \cos nx + b_n \sin nx \right)^2 dx$$

$$= \int_{-\pi}^{\pi} a_0^2 dx + \sum_{i=1}^{\mathrm{N}} \left(a_n^2 \int_{-\pi}^{\pi} \cos^2 nx dx + b_n^2 \int_{-\pi}^{\pi} \sin^2 nx dx + \int_{-\pi}^{\pi} \cos nx \sin mx \, dx \right)$$

$$= 2\pi a_0^2 + \sum_{i=1}^{\mathrm{N}} (\pi a_n^2 + \pi b_n^2)$$

$$\mathrm{E} = \int_{-\pi}^{\pi} (f^2 - 2f\mathrm{F} + \mathrm{F}^2) \, dx$$

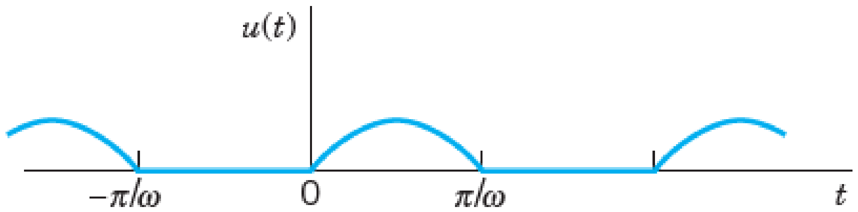
$$= \int_{-\pi}^{\pi} f^2 \; dx - 4\pi a_0^2 - 2\pi \sum_{i=1}^{\mathrm{N}} (a_n^2 + b_n^2) + 2\pi a_0^2 + \pi \sum_{i=1}^{\mathrm{N}} (a_n^2 + b_n^2)$$

$$\boxed{\mathrm{E} = \int_{-\pi}^{\pi} f^2 \, dx \; - \; \pi \left[2a_0^2 + \sum_{n=1}^{\mathrm{N}} (a_n^2 + b_n^2) \right]}$$

Practice Problems

Problem 1: Find the Fourier series representation of $f(x)$

$$f(x) = \begin{cases} 0, & if\ -\mathrm{L} < t < 0 \\ \mathrm{E}\sin\omega t\,, & if\ 0 < t < \mathrm{L} \end{cases}$$

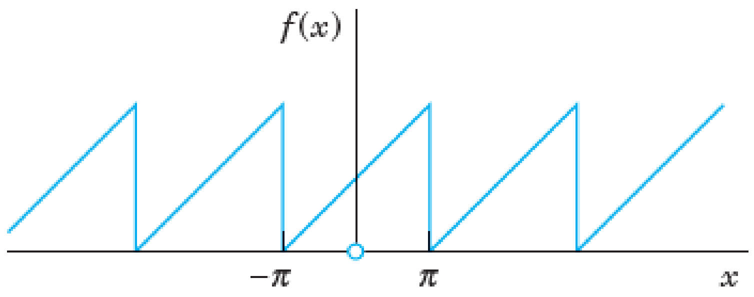


Half-wave rectifier

$$p = 2\mathrm{L} = \frac{2\pi}{\omega}$$

Problem 2: Find the Fourier series representation of $f(x)$

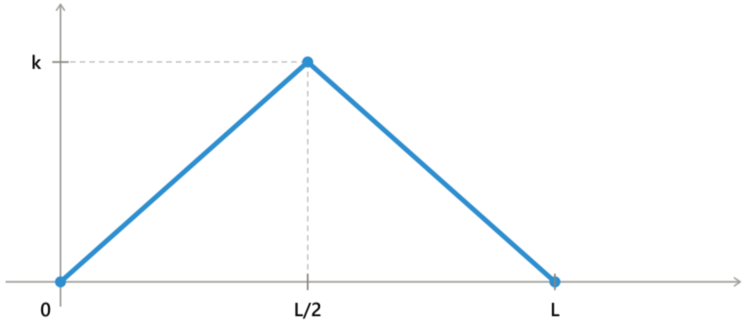
$$f(x) = x + \pi, \qquad if\ -\pi < x < \pi$$



Sawtooth Wave

Problem 3: Find the even and odd extension of $f(x)$

$$f(x)=\begin{cases}\dfrac{2k}{\mathrm{L}}x, & if\ 0<x<\dfrac{\mathrm{L}}{2}\\ \dfrac{2k}{\mathrm{L}}(\mathrm{L}-x), & if\ \dfrac{\mathrm{L}}{2}<x<\mathrm{L}\end{cases}$$



Orthogonal Functions

» Orthogonal functions

- Functions $y_1(x), y_2(x), \ldots$ defined on some $a \le x \le b$ are called orthogonal with respect to the weight function, $r(x) > 0$ if

$$(y_m, y_n) = \int_a^b r(x)\, y_m(x) y_n(x) dx = 0, \qquad (m \ne n)$$

If $r(x) = 1$

$$(y_m, y_n) = \int_a^b y_m(x) y_n(x) dx = 0, \qquad (m \ne n)$$

- Norm of functions $y_1(x), y_2(x), \ldots$ defined on some $a \le x \le b$

$$\|y_m\| = \sqrt{(y_m, y_m)} = \sqrt{\int_a^b r(x)\, y_m^2(x) dx}$$

- Orthonormal functions $y_1(x), y_2(x), \ldots$

$$(y_m, y_n) = \int_a^b r(x)\, y_m(x) y_n(x) dx = \delta_{mn} = \begin{cases} 0, & if\ m \ne n \\ 1, & if\ m = n \end{cases}$$

Orthogonal Functions

» Orthogonal functions

$$y_m(x) = \sin mx, \qquad m = 1,2,\ldots \qquad -\pi \le x \le \pi$$

$$y_n(x) = \sin nx, \qquad r(x) = 1$$

$$(y_m, y_n) = \int_{-\pi}^{\pi} \sin mx \sin nx \, dx, \qquad (m \neq n)$$

$$= 0$$

$$\|y_m\| = \sqrt{(y_m, y_m)}$$

$$(y_m, y_m) = \int_{-\pi}^{\pi} \sin^2 mx \, dx = \frac{1}{2}\int_{-\pi}^{\pi} (1 + \cos 2mx) \, dx$$

$$= \frac{1}{2}\left[\int_{-\pi}^{\pi} dx + \int_{-\pi}^{\pi} \cos 2nx \, dx\right]$$

$$= \frac{1}{2}\left[(2\pi) + \frac{\sin 2nx}{2n}\bigg|_{-\pi}^{\pi}\right]$$

$$= \pi$$

» Orthonormal functions

$$y_m(x) = \frac{\sin mx}{\sqrt{\pi}}, \qquad m = 1,2,\ldots \qquad -\pi \le x \le \pi$$

» Sturm – Liouville Problem

- Sturm – Liouville equation

$$[p(x)y']' + [q(x) + \lambda r(x)]y = 0$$

- Boundary conditions

$$k_1 y + k_2 y' = 0, \quad \text{at } x = a$$

$$l_1 y + l_2 y' = 0, \quad \text{at } x = b$$

- $y = 0$ is the trivial solution of the Sturm – Liouville problem.
- The non – trivial solutions, $y(x)$ are known as eigenfunctions, and the number λ for which an eigenfunctions exists an eigenvalue of the Sturm – Liouville problem.

Singular Sturm – Liouville problem

$p(a) = 0$, or $p(b) = 0$

$a \to \pm\infty$, or $b \to \pm\infty$

Periodic Sturm – Liouville problem

$y(a) = y(b)$

$y'(a) = y'(b)$

Sturm – Liouville Problem

» Sturm – Liouville Problem: Eigen functions

$$[p(x)y']' + [q(x) + \lambda r(x)]y = 0$$

- p, q, r, p' in the Sturm – Liouville equation are real-valued and continuous
- Let $y_m(x)$ and $y_n(x)$ be the eigenfunctions of the Sturm – Liouville problem corresponding to different eigenvalues λ_m and λ_n, respectively.
- $y_m(x)$ and $y_n(x)$ are orthogonal with respect to weight function, $r(x)$

$$(y_m, y_n) = \int_a^b r(x)\, y_m(x) y_n(x) dx = 0, \qquad (m \neq n)$$

» Example: Legendre's equation

$$(1 - x^2)y'' - 2xy' + n(n+1)y = 0 \qquad -1 \leq x \leq 1$$

$$[(1 - x^2)y']' + n(n+1)y = 0$$

$$[(1 - x^2)y']' + \lambda y = 0, \qquad \lambda = n(n+1)$$

Sturm – Liouville Problem

$$[(1 - x^2)y']' + \lambda y = 0, \qquad \lambda = n(n+1)$$

Compared it with Sturm – Liouville equation, we get

$$p(x) = 1 - x^2, \qquad q(x) = 0, \qquad r(x) = 1$$

$p(-1) = 0$, $p(1) = 0$, the Sturm – Liouville problem is singular.

$$\mathrm{P}_1(x) = 1$$

$$\mathrm{P}_2(x) = x$$

$$\mathrm{P}_3(x) = \frac{1}{2}(3x^2 - 1)$$

$$\mathrm{P}_4(x) = \frac{1}{2}(5x^3 - 3x)$$

Legendre’s Polynomials

$$\int_{-1}^{1} \mathrm{P}_1(x)\mathrm{P}_2(x)dx = \int_{-1}^{1} 1.xdx = \int_{-1}^{1} xdx = \left[\frac{x^2}{2}\right]_{-1}^{1} = 0$$

$\mathrm{P}_1(x)$ and $\mathrm{P}_2(x)$ are orthogonal polynomials

Sturm – Liouville Problem

» Example: Bessel's equation

$x^2y'' + xy' + (x^2 - n^2)y = 0 \qquad\qquad 0 \le x \le a$

When studying Bessel functions as Sturm – Liouville problem, we generally write

$$x^2y'' + xy' + (k^2x^2 - n^2)y = 0$$

$$xy'' + y' + \left(k^2x - \frac{n^2}{x}\right)y = 0$$

$$[xy']' + \left(k^2x - \frac{n^2}{x}\right)y = 0$$

$$[xy']' + \left(\lambda x - \frac{n^2}{x}\right)y = 0$$

Compared it with Sturm – Liouville equation, we get

$$p(x) = x, \qquad q(x) = -\frac{n^2}{x}, \qquad r(x) = x$$

Bessel function of the first kind, $\mathrm{J}_n(kx)$ forms an orthogonal set

Apply BC $y\,(a) = 0$, we get

$\mathrm{J}_n(ka) = 0$ which determines the eigen values $\lambda_1, \lambda_2, \ldots$

Generalized Fourier Series

Generalized Fourier series

- Functions $y_1(x), y_2(x), \ldots$ defined on some $a \le x \le b$ be orthogonal with respect to the weight function, $r(x) > 0$

$$f(x) = \sum_{m=0}^{\infty} a_m y_m(x) = a_0 y_0(x) + a_1 y_1(x) + \cdots$$

$$a_m = \frac{(f, y_m)}{\|y_m\|^2} = \frac{1}{\|y_m\|^2} \int_a^b r(x) f(x) y_m(x) dx, \qquad n = 0,1, \ldots$$

This is called orthogonal series, orthogonal expansion, or generalized Fourier series.

- If the $y_m(x)$ are the eigenfunctions of Sturm-Liouville problem, then the generalized Fourier series is termed as eigenfunctions expansion.

Generalized Fourier Series

» Example: Fourier – Legendre series

$$f(x) = \sum_{m=1}^{\infty} a_m \mathrm{P}_m(x) = a_0\mathrm{P}_0 + \ a_1\mathrm{P}_1(x) + a_2\mathrm{P}_2(x) + \ \dots$$

$\mathrm{P}_0, \mathrm{P}_1(x), \mathrm{P}_2(x)$ are Legendre polynomials

$$a_m = \frac{1}{\|\mathrm{P}_m\|^2}\int_{-1}^{1} f(x)\mathrm{P}_m(x)dx\,, \qquad n = 0,1,\dots$$

$$\|\mathrm{P}_m\| = \sqrt{\int_{-1}^{1} \mathrm{P}_m^2(x)dx} = \sqrt{\frac{2}{2n+1}}, \qquad n = 0,1,\dots$$

Let us try to approximate the below function

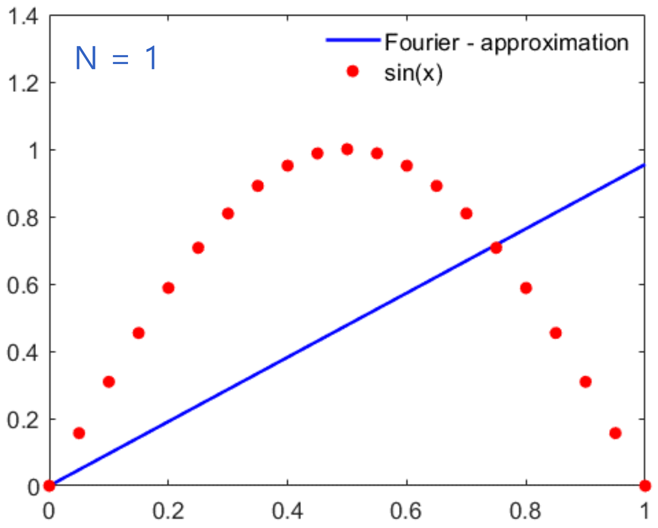
$$f(x) = \sin \pi x$$

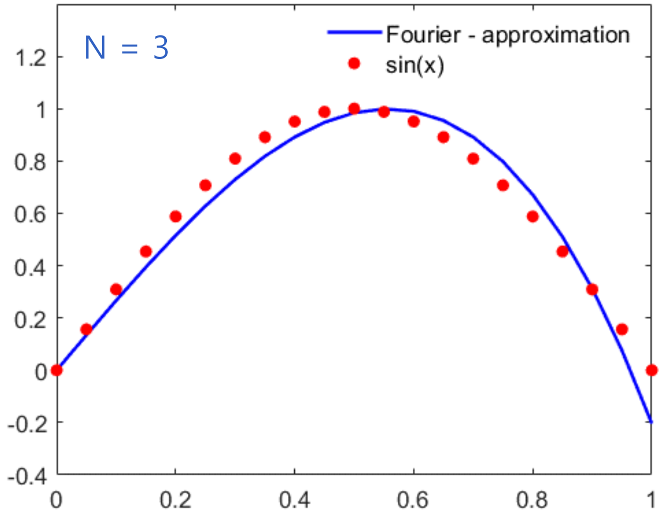
$$a_m = \frac{2n+\ 1}{2}\int_{-1}^{1} \sin \pi x\, \mathrm{P}_m(x)dx\,, \qquad n = 0,1,\dots$$

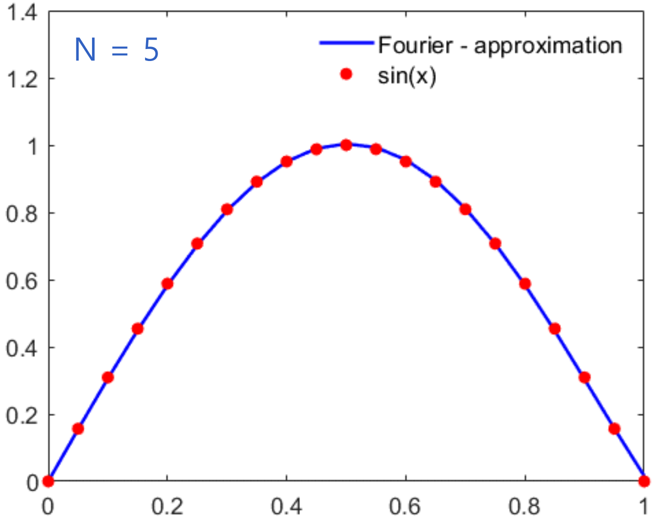
Generalized Fourier Series

```matlab
% Approximation of sin(x) using Fourier - Legendre series
% SYMBOLIC MATH TOOLBOX
syms x
t = 0:0.05:1;
s = zeros(1, size(t, 2));
for n = 1:5
    a = legpol(n);
    f = double(a*subs(legendreP(n,x),x,t));
    s = s + f;
end
plot(t,s,t,sin(pi*t),'o')
```

```matlab
function a = legpol(n)
% Calculate the Fourier - Coefficient
syms x
P = legendreP(n,x);
f = sin(pi*x)*P;
a = 0.5*(2*n+1)*int(f,-1,1);
a = double(a);
end
```







Generalized Fourier Series

» Convergence

- A sequence of functions, f_k is called convergent with the limit f if

$$\lim_{k\to\infty} \|f_k - f\| = 0$$

- For the generalized Fourier series

$$f_k(x) = \sum_{m=0}^{k} a_m y_m(x)$$

In the mean – squared error sense

$$\lim_{k\to\infty} \int_a^b r(x)[f_k(x) - f(x)]^2 dx = 0$$

» Bessel's inequality

$$\boxed{\sum_{m=0}^{k} a_m^2 \le \|f\|^2 = \int_a^b r(x)[f(x)]^2 dx, \qquad k = 1, 2, \ldots}$$

Generalized Fourier Series

» Bessel's inequality: Proof

$$\int_a^b r(x)[f_k(x) - f(x)]^2 dx = \int_a^b (rf_k^2 \; - 2rff_k + rf^2)dx \;\; = \int_a^b rf_k^2 dx - 2\int_a^b rff_k dx + \int_a^b rf^2 dx$$

$$\therefore \int_{-\pi}^{\pi} rf_k^2 dx = \int_{-\pi}^{\pi} r\left(\sum_{m=0}^{k} a_m y_m(x)\right)^2 dx$$

$$= \sum_{m=0}^{k} a_m^2 \int_a^b r y_m y_n dx + \sum_{m=0}^{k} a_m^2 \int_a^b r y_m^2 dx$$

$$= \sum_{m=0}^{k} a_m^2 \|y_m\|^2$$

$$= \sum_{m=0}^{k} a_m^2$$

$$\therefore \int_a^b rff_k dx = \int_{-\pi}^{\pi} rf \sum_{m=0}^{k} a_m y_m(x)\, dx$$

$$= \sum_{m=0}^{k} a_m \int_a^b rf y_m dx$$

$$= \sum_{m=0}^{k} a_m \, a_m \|y_m\|^2$$

$$= \sum_{m=0}^{k} a_m^2$$

Generalized Fourier Series

$$\therefore \int_a^b r(x)[f_k(x) - f(x)]^2 dx = \sum_{m=0}^{k} a_m^2 - 2\sum_{m=0}^{k} a_m^2 + \int_a^b r f^2 dx$$

$$= -\sum_{m=0}^{k} a_m^2 + \int_a^b r f^2 dx \geq 0$$

$$\sum_{m=0}^{k} a_m^2 \leq \|f\|^2 = \int_a^b r(x)[f(x)]^2 dx, \qquad k = 1, 2, \dots$$

- As $k \to \infty$, the partial sums are increasing and bounded, therefore

$$\sum_{m=0}^{\infty} a_m^2 \leq \|f\|^2$$

Generalized Fourier Series

» Completeness

An orthonormal set $y_0(x), y_1(x), y_2(x), \dots$ on an interval $a \le x \le b$, is complete in a set of functions S defined on $a \le x \le b$ if we can approximate every f belonging to S arbitrarily close in the norm by a linear combination of the form

$$\|f - (a_0 y_0 + \dots + a_k y_k\| < \in$$

- As $k \to \infty$, for a particular $f \in$ S,

  if f cannot be represented completely by the orthonormal set $y_0(x), y_1(x), y_2(x), \dots$, then

$$\sum_{m=0}^{\infty} a_m^2 < \|f\|^2$$

» Parseval's equality

$$\sum_{m=0}^{\infty} a_m^2 = \|f\|^2 = \int_a^b r(x)[f(x)]^2 dx$$

Parseval's equality is the mathematical statement of energy/norm invariance under an orthonormal Fourier transformation.

Practice Problem

Problem 1: Find the eigenvalues and eigenfunctions of the Sturm–Liouville problem

$$y'' + \lambda y = 0, \qquad y(0) = 0, y(\pi) = 0$$